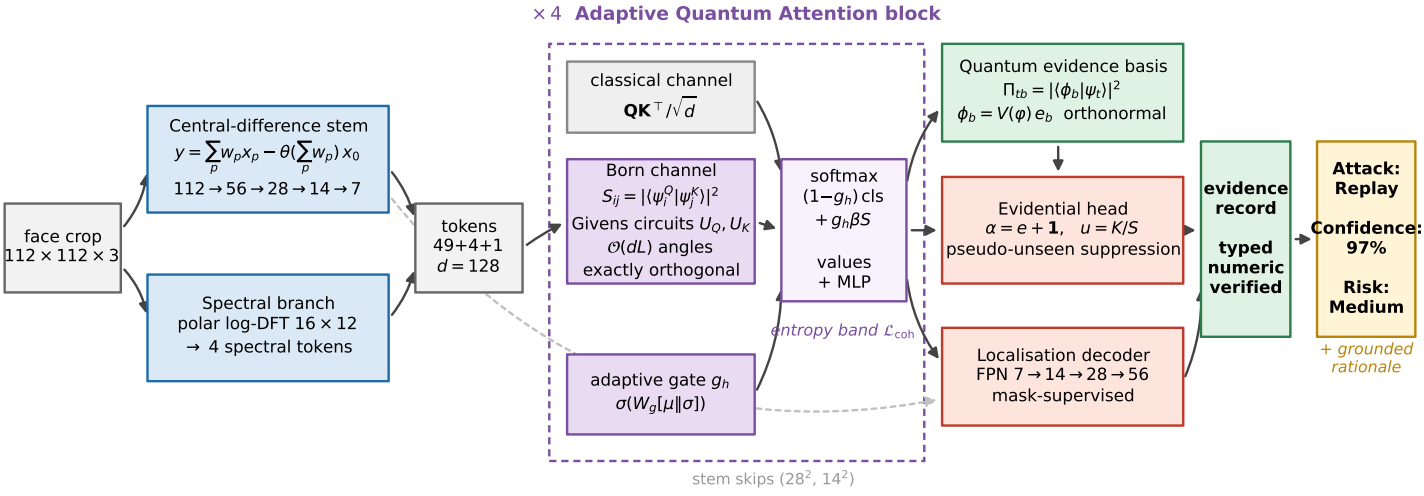

× 4 Adaptive Quantum Attention block
face crop
112 × 112 × 3
Central-difference stem
$y = \sum_p w_p x_p - \theta(\sum_p w_p)\, x_0$
112 → 56 → 28 → 14 → 7
Spectral branch
polar log-DFT 16 × 12
→ 4 spectral tokens
tokens
49+4+1
d = 128
classical channel
$\mathbf{QK}^\top/\sqrt{d}$
Born channel
$S_{ij} = |\langle\psi_i^Q|\psi_j^K\rangle|^2$
Givens circuits U_Q, U_K
$\mathcal{O}(dL)$ angles
exactly orthogonal
adaptive gate g_h
$\sigma(W_g[\mu\|\sigma])$
softmax
$(1-g_h)$ cls
$+ g_h \beta S$
values
+ MLP
entropy band $\mathcal{L}_{\text{coh}}$
stem skips (28^2, 14^2)
Quantum evidence basis
$\Pi_{tb} = |\langle\phi_b|\psi_t\rangle|^2$
$\phi_b = V(\varphi)\, e_b$ orthonormal
Evidential head
$\alpha = e + \mathbf{1}$, $u = K/S$
pseudo-unseen suppression
Localisation decoder
FPN 7 → 14 → 28 → 56
mask-supervised
evidence record
typed
numeric
verified
Attack: Replay
Confidence: 97%
Risk: Medium
+ grounded rationale